

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

grn_interaction_pattern_mapping.py

By this point there are two separate pictures: the GRN says which transcription factors are behind the mutant changes, and the interaction model says which genes switch on their problem with age. The script asks whether those are the same picture, by taking the genes the GRN's TFs control and looking up how they behave across age.

- **grn_target_interaction_map.csv:** the bridge between the two analyses, one row per GRN target gene. It carries the zebrafish gene, its human symbol, how many of the GRN transcription factors regulate it, and its full interaction behaviour: its pattern, its direction in the mutant, and its interaction fold change and q-value. For any target gene we can now read in one place both which switches control it and how it changes with age.
- **grn_tf_pattern_summary.csv:** one row per transcription factor, giving its activity score, how many of its targets could be mapped, and the percentage of those targets that are age-dependent onset, that go down in the mutant, and that have a significant interaction. This is the table where E2F stands out. Around 76 to 91 percent of the E2F targets are both switched off in the mutant and age-dependent, which is the tightest link between the two analyses anywhere in the project.
- **grn_pattern_enrichment.png:** bar charts comparing the GRN target genes against the rest of the genome. 68.3 percent of GRN targets show age-dependent onset compared with 34.6 percent of genes generally, which is a 4.8 times enrichment and statistically overwhelming. In plain terms, the transcription factors identified by the GRN are specifically controlling the programme that switches on with age, which is what ties the two halves of the project together.